

Week D - By Deng Yufan.

Prob 1. Proof $\forall$ AM for lang. A , Arthur runs k independent msg. and accept iff all k copies accept. then accept $x \notin A$ prob. at most 2^{-k} . Merlin's msg. may be not independent.

Say a msg. r sent by A is bad on $x \notin A$. if $\exists s$ that $V(x, r, s) = 1$. Since if r is bad, Merlin can compute s , and reply to Arthur, so $P(r \text{ is bad}) \leq \frac{1}{2}$.

Now we run k copies. $P(\text{Arthur accept}) = P(\bigwedge_i \text{Arthur accept } (r_i, s_i))$

$$\leq P(\bigwedge_i r_i \text{ is bad})$$

$$= P(r \text{ is bad})^k = 2^{-k}.$$

Prob 2.

a. Explain the following $S(x)$ ~~doesn't~~ doesn't work in PZK proof for GRAPH-ISO protocol discussed in class: 1. choose $a \in \{0, 1\}$. 2. choose π . compute $G = \pi(G_a)$.
(or write below) 3. choose $b \in \{0, 1\}$. 4. if $b = a$. output. otherwise $\bullet$ rewind.

For V^* that not always output random bit, e.g. $V^* = \text{output } 0$, $S(x) \neq \text{view}_{V^*} \langle P, V^* \rangle(x)$, since the distribution for b in ~~is~~ is different. V^* is always 0, but S is $\frac{1}{2} 0, \frac{1}{2} 1$. (G, b, G).

b. Prove the PZK for GRAPH-ISO protocol: 1. P randomly perm. G_0 into H by σ .

2. V rand pick $b \in \{0, 1\}$.

3. P give witness for $H \cup G_b$ (a perm.)

and S : 1. rand pick $a \in \{0, 1\}$. random perm. G_a into H by σ .

2. run V^* with H . get $b \in \{0, 1\}$.

3. if $a = b$. output (H, b, σ) . otherwise rewind.

Suppose

~~For~~ $x \in A$, or say $G_0 \cup G_1$. $\forall V^* = V^*(x, H, \text{randomness } r) \in \{0, 1\}$, the distribution of $\text{view}_{V^*} \langle P, V^* \rangle(x)$ is: H is uniform random perm. of G_0 . ~~and~~
 $b = V^*(x, H)$.

σ is a witness for $H \cup G_b$.

For S : H is a random perm. of random G_0 or G_1 . Since $G_0 \cup G_1$, suppose $G_1 = \pi(G_0)$, $H = \sigma(G_0)$ or $\sigma(\pi(G_0))$. Since distribution for σ and $\sigma \circ \pi$ is the same (both random uniformly perm), H is equiv to $\bullet$ uniform random perm. of G_0 . (Rewind only change distribution for a , not for H)

Obviously $b = V^*(x, H)$. and σ is a witness for $H \cup G_b$ since $H = \sigma(G_a) = \sigma(G_b)$.

So $S(x) = \text{view}_{V^*} \langle P, V^* \rangle(x)$. thus the protocol is PZK.

